

Aryavardhan Singh Shaktawat

☎ +91 9950634068 📁 portfolio in linkedin 🌐 aryavardhan007 ✉ aryavardhanshaktawat99@gmail.com

Education

Indian Institute of Technology Jodhpur
B.Tech in Electrical Engineering

June 2018 – May 2022

Experience

Avataar.ai - Senior AI Software Engineer

Jan 2026 - Present — Bangalore, India

Kisan-Vegh & UAI Dashboard — Voice-First AI Farming Assistant

- Architected a **production-grade voice-first AI farming assistant** with end-to-end flow across **real-time audio transcription**, multimodal image analysis, and multilingual LLM responses.
- Built a **multi-model orchestration pipeline** (Sarvam ASR → Dhenu VLM → LLM) with centralized error handling, per-model fallbacks, and graceful degradation.
- Delivered a **real-time multi-agent observability dashboard** (Socket.io + custom React hooks) with event-driven state transitions and sub-second UI updates; **presented at Impact AI Summit 2026**.

Avataar Flows — AI Workflow Automation Platform

- Engineered a **visual workflow automation platform** (React 18 + TypeScript) with a **drag-and-drop canvas editor** (React Flow + Dagre), real-time execution monitoring, and **step-by-step HITL approvals** across 5 action types.
- Built a **multi-tenant AI workflow execution engine** (FastAPI + PostgreSQL + asyncio) supporting AGENT, HUMAN, and SYSTEM node types with **DAG-based parallel execution** and a **thread-pool executor** (16 workers, queue of 300) for non-blocking concurrency.
- Implemented **Tattva** — a semantic entity layer with **pgvector embeddings** (Voyage AI, 1024-dim), **multi-provider LLM support** (OpenAI, Anthropic, Gemini), and a one-click domain template system — achieving **89% accuracy at 33% of GPT-4o inference cost**.
- Deployed on **AWS EKS (Kubernetes)** with CI/CD via Jenkins → ECR → rolling deploy; integrated **AWS S3, Cognito, API Gateway, CloudFront, and SES** across frontend and backend services.

UAI — Unified AI Interface (National-Scale AI Protocol)

- Built three microservices (**Registry, Trust Server, Audit Ledger**) using **FastAPI + PostgreSQL + Elasticsearch + Redis**, forming the control plane for a **national-scale AI protocol** with **append-only audit storage** and SHA-256 receipt hashing.
- Implemented **W3C DID v1.0** resolution, **Ed25519/P-256** signature verification, **W3C Verifiable Credential** validation, and **JWT delegation tokens** with risk-tiered TTLs — with full **DPDP Act 2023** and **CERT-In** compliance (TLS 1.3, 180-day log retention).
- Co-designed a **5-layer protocol architecture** with bridge adapters for **MCP, Beckn, and A2A** interoperability; set up Docker Compose dev infra, **CI/CD conformance test harness**, and **Prometheus + Grafana** monitoring.

Enrich Beauty - Senior Software Developer

Nov 2022 - Jan 2026 — Bangalore, India

- Architected a **revenue synchronization system** (Firebase Functions + BigQuery) processing invoices, credit notes, and orders with **automated failure handling, dead-letter logic, and retry mechanisms**, achieving **99%+ synchronization accuracy**.
- Designed a **vendor management platform** with event-driven **BigQuery** sync and **automated MRP recalculation** via Pub/Sub; architected an **end-to-end GST-compliant e-invoicing and e-way bill system** with **150+ regulatory checks**, reducing warehouse manual effort by **95%+**.
- Built a **configuration-driven fortune wheel system** (Flutter + Firebase Functions) with **Firestore transactions** for atomic reward allocation — driving **1.5× engagement** and **+5% avg. order value**; implemented a **local-first ISAR cache** with TTL-based invalidation, reducing Firestore reads by **80%**.

Reliance Jio Platforms - Software Development Engineer

July 2022 - Nov 2022 — Mumbai, India

- Built **scalable microservice integrations** using **Java and Spring Boot**, leveraging **Redis Sentinel** and **Azure Service Bus** to improve the performance, throughput and reliability of WhatsApp bot workflows.

Technical Skills

Languages: Python, TypeScript, Dart, C++, SQL, HTML, CSS

AI/ML & LLM Systems: LLM Orchestration, Agent Workflows, RAG Pipelines, Vector Search (pgvector), Embeddings (Voyage AI), Multi-LLM Integration, Context Assembly, Semantic Retrieval

Frontend: React 18, React Flow, Zustand, TanStack Query, Axios, Flutter, Vite, SCSS Modules

Backend: FastAPI, Django, Node.js, Spring Boot, PostgreSQL, Redis, Elasticsearch, Celery, asyncio, SQLAlchemy

Cloud/DevOps: AWS (EKS, S3, Cognito, API Gateway, CloudFront, SES, ECR), GCP, Firebase, Docker, Kubernetes, Jenkins, Prometheus, Grafana, kibana

Standards/Security: W3C DID, Verifiable Credentials, JWT (RFC 7519), Ed25519, TLS 1.3, DPDP Act 2023, CERT-In, MCP, Beckn Protocol

Achievements

- Won Taking Ownership and Outstanding Delivery at Enrich for Successfully developing 5+ high impact projects.
- Student Guide at Counselling Service (IIT Jodhpur) – Guided and mentored 8 freshmen in their freshman year.